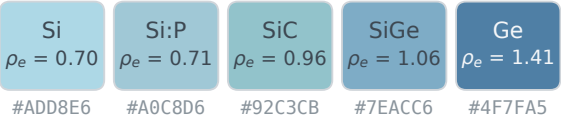


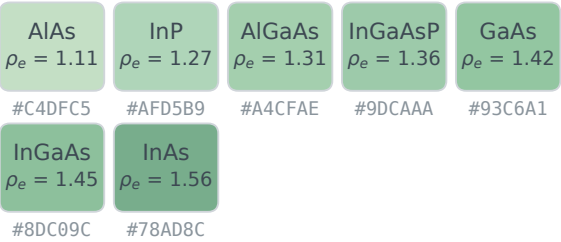
Material colors for cross-section schematics

- 1 hue = chemical family
- 2 lightness = electron density ρ_e (heavier $\rightarrow$ darker)
- 3 alloys interpolate their endpoint colors
- 4 doped variants retain the host hue and darken only when they must be distinguished

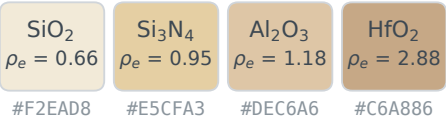
Group IV – blues (SiC = silicon carbide)



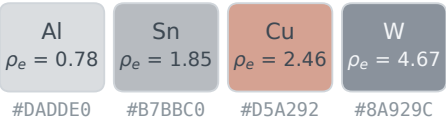
III-V – greens



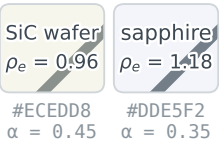
dielectrics – soft sands



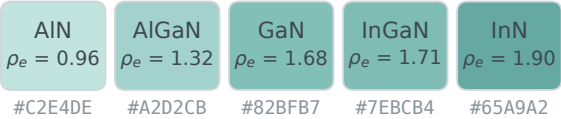
metals – greys and native tints



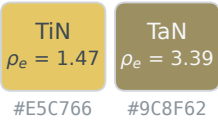
transparent crystals – near-clear, translucent



III-nitrides – teals



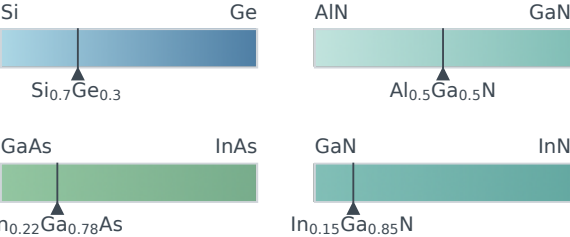
barrier nitrides – metallic gold-olive



neutrals and packaging



alloys interpolate endpoint colors



ρ_e : reference electron density ($e \text{ \AA}^{-3}$) from bulk density; higher values map to darker fills.
Substrate, package and PCB are semantic materials and take documented colors rather than a ρ_e shade.
Transparent crystals (bulk SiC wafer, sapphire = Al₂O₃) are near-clear tints drawn at the stated α by every renderer; the stripe shows the transparency. They are distinct from the opaque device color SiC and the thin-film dielectric Al₂O₃.
Alloy ramps: x is the right-hand endpoint fraction; RGB interpolation is a drawing convention, not a physical model.